

The Fundamental Theorem of Coherent Residual Pathways: The Generalized Q-LINK Ansatz

Satya Pratheek Tata

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Abstract

We define the absolute boundaries of the coherent residual messenger architecture by identifying its irreducible algebraic invariants versus its flexible hyperparameters. Synthesizing the topological, temporal, and spatial constraints derived in Extensions 1 through 11, we formulate the “Generalized Residual Ansatz.” This formulation establishes the most general class of parameterized quantum circuits that mathematically guarantee a volumetric barren plateau mitigation advantage over Vanilla architectures, defining the strict prerequisites for Dynamical Lie Algebra (DLA) expansion and Hilbert-Schmidt trace asymmetry.

1 The Irreducible Invariants

Any architecture seeking to exploit the Q-LINK trace advantage must strictly obey three geometric and algebraic invariants. Violating any of these triggers an absorbing state (exponential volume penalty, DLA fracturing, or directional collapse).

1. **The Dimensional Constraint (The Volume Axiom):** The messenger register must be exactly $k = 1$. The extended Hilbert space is strictly $\mathcal{H}_{tot} = \mathcal{H}_{data} \otimes \mathbb{C}^2$. Any $k > 1$ introduces an $\mathcal{O}(2^{-k})$ Haar measure suppression that strictly overpowers linear trace advantages.
2. **The Trace Asymmetry (The Measurement Axiom):** The cost operator O must act trivially on the messenger qubit: $\tilde{O} = O_{data} \otimes I_m$. If the messenger is measured, the Hilbert-Schmidt trace symmetry is restored, and the absolute variance ratio collapses to 1.
3. **The Orthogonal Hinge (The Commutator Axiom):** The collection layer and distribution layer must project strictly anti-commuting Pauli operators onto the messenger subspace. If $G_{coll} \propto Q_m \otimes A_i$ and $G_{dist} \propto P_m \otimes B_i$, then $\{Q_m, P_m\} = 0$. Furthermore, the distribution generator must utilize the messenger explicitly as the control (P_m), not the target, to prevent directional DLA collapse.

2 The Generalized Residual Ansatz

Provided the invariants in Section 1 are maintained, all other structural components of the network are completely elastic. We define the Most General Form of the architecture as the tuple $\mathcal{A} = (\mathcal{G}_{opr}, Q, P, p, m)$:

- $\mathcal{G}_{opr}$: Any universal set of generators on $\mathcal{H}_{data}$ that spans $\mathfrak{su}(2^n)$.
- $Q, P \in \{X, Y, Z\}$: The messenger interaction bases, subject to $\{Q, P\} = 0$.

- $p \in \mathbb{N}^+$: The period of the residual layer application (temporal sparsity).
- $m \in [1, n]$: The locality of the normalized averaged cost operator on $\mathcal{H}_{data}$.

The general unitary for a single periodic block (spanning p layers) is:

$$U_{block}(\theta) = U_{dist}^{(P)} U_{coll}^{(Q)}(\theta) \prod_{l=1}^p U_{opr}^{(l)}(\theta) \quad (1)$$

Where the residual boundary layers are defined by their arbitrary non-commuting generators:

$$U_{coll}^{(Q)} = \exp \left(-i \sum_{i=1}^n \alpha_i (Q_m \otimes A_i) \right), \quad U_{dist}^{(P)} = \exp \left(-i \sum_{i=1}^n \beta_i (P_m \otimes B_i) \right) \quad (2)$$

3 The Universal Variance Bound

Because the orthogonal hinge $\{Q, P\} = 0$ guarantees that the Lie brackets successfully break out of any single Pauli eigenbasis, the iterated closure of the Generalized Ansatz strictly generates the full extended special unitary group:

$$\mathfrak{g}_{\mathcal{A}} = \langle \mathcal{G}_{opr} \cup \{Q_m \otimes A_i\} \cup \{P_m \otimes B_i\} \rangle_{Lie} = \mathfrak{su}(2^{n+1}) \quad (3)$$

Because all valid variations of $\mathcal{A}$ eventually saturate the identical $U(2^{n+1})$ Haar measure, the Master Formula denominator is invariant. The absolute trainability advantage depends entirely on the spatial locality m shielding the trace numerator.

The Fundamental Theorem: Any quantum neural network obeying the Generalized Residual Ansatz $\mathcal{A}$ will exhibit an asymptotic gradient variance ratio relative to an equivalent $(n+1)$ -qubit Vanilla architecture of exactly:

$$\mathcal{R}_{\mathcal{A}}(n) = \frac{n+1}{n+1-m} \quad (4)$$

This bound is mathematically absolute, globally optimal, and perfectly agnostic to the specific choice of hardware-native entanglers (P, Q) and finite-depth optimization throttling (p) .